

Battleshots | TrustGuard technical report

2026-10-04 | Track 2: Intelligent Financial Fraud Detection

1. Problem Understanding

We rank transactions by fraud risk from supplied historical labels. Fraud appears in 353 of 20,000 training rows (1.765%). Calling every row legitimate gives 98.235% accuracy but catches no fraud. PR-AUC (average precision) evaluates how well the score ranks fraud across thresholds. A yes/no decision needs a separate review-capacity and false-alarm tradeoff.

2. Data Processing & Feature Engineering

The model never uses transaction ID or the fraud label as an input. It caps recent 24-hour spend at the labelled-training 99th percentile (4,926.9294) to limit extrapolation. Its 32 inputs include seven numeric fields, three categorical fields, logs, recent-spend ratios, night/new-account/foreign flags, missingness flags and device interactions. The new interaction marks night transactions on a new device. Numeric medians, scaling and one-hot categories are fitted inside each training fold; unseen categories are accepted.

3. Model Selection & Justification

Controlled comparisons of logistic, tree, spline, nonlinear and ranking models did not give a stronger confirmed model. A risk audit found 67 frauds among 397 labelled night/new-device rows; the prior model underpredicted this segment's total. Adding one interaction to regularized logistic regression ($C=0.3$) improved all five validation seeds. The original +0.010 mean-gain gates were missed narrowly, so we made one explicitly exploratory public probe. CSV #3965 scored **0.19095** versus prior #3683 at 0.18568. Public selection is optimistic; no private gain is established.

4. Evaluation Strategy

Five-fold stratified validation on five seeds, with fold-local preprocessing, gave mean PR-AUC **0.231223** versus **0.221725** for the prior model (gain 0.009498). The first-seed gain was 0.007764; a paired bootstrap 95% interval crossed zero [-0.00216, 0.01661]. Training on the lowest 16,000 numeric IDs and evaluating the highest 4,000 gave 0.234311 versus 0.221582; ID order is not proven chronological. In the first out-of-fold run, the top 1% contains 79/353 frauds (39.5% precision) and top 3% contains 116/353 (19.3%). These are labelled-validation results, not hidden-test counts.

5. Limitations & Risks

Only 353 labelled frauds make small metric changes uncertain. The test distribution shifts: 90th-percentile amount rises from 530 to 1,707, and rows with any missing field rise from 4.9% to 13.9%. A feature-only test-like holdout favored the new model, but an earlier proxy failed to predict public performance, so it is weak evidence. The public leaderboard and repeated model selection can overstate private performance; the private score is unknown. Risk scores are not proof of fraud or guaranteed calibrated probabilities. The saved JSON scorer and prediction-only notebook reproduce all 12,000 public scores on shuffled, ID-free input; v4 passed the live sandbox and wrote 20,000 scores.